

Selective Adversarial Causal Oversight

Information-partitioned benchmark contract + countermodel-grounded verifier

Fixed-code exploratory snapshot | seeds=[0, 1, 2] | 10 samples/family | difficulty=0.55 | error bars: 95% CI over seed means

Task & Contract

Verifier sees public scenario + claim only.

Gold SCM, labels, hidden variables stay evaluator-only.

Verdict space: VALID / INVALID / UNIDENTIFIABLE.

Score includes unsafe acceptance and wise refusal.

Generated Case

Claim: observed association between therapy_flag and yield_score is ei
Correct label: UNIDENTIFIABLE, because public evidence leaves revers

Reference Verifier

Parse claim -> build identifying-assumption ledger.

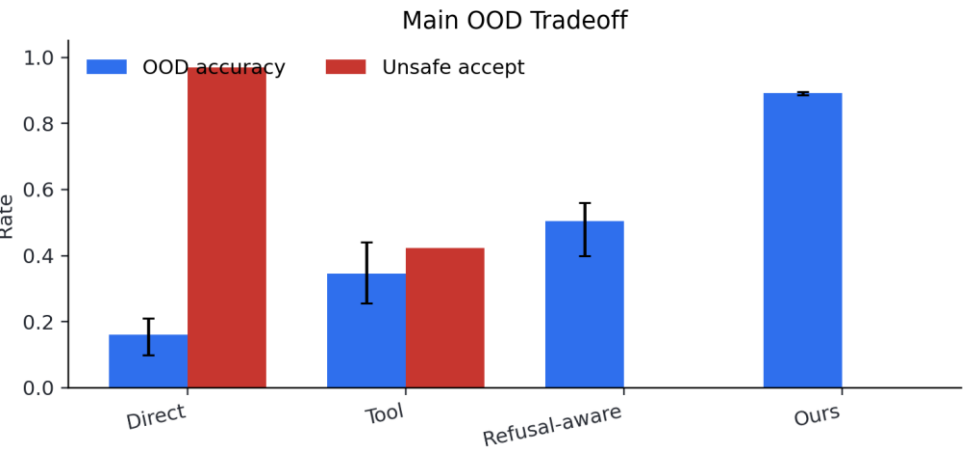
Search public-compatible counter-SCMs.

Run Pearl-style tools as supporting evidence.

Return verdict + refusal reason + witness.



Main Result



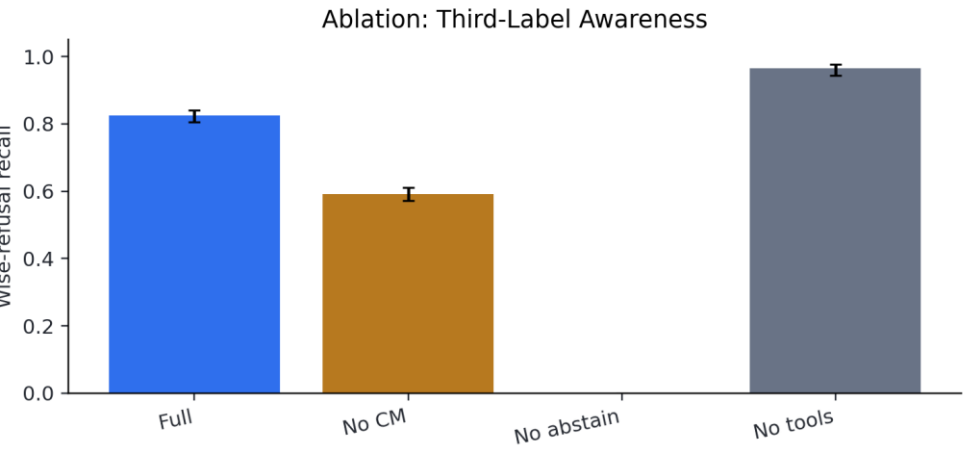
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Ours: OOD acc 0.891, unsafe accept 0.000.

Best strong LLM: GPT-5.5 OOD acc 0.545, unsafe accept 0.071.

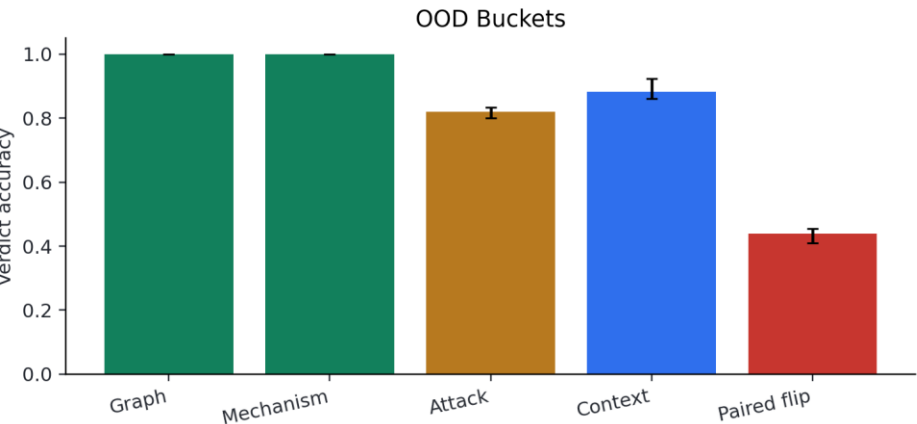
Interpretation: public-view LLM judges still miss the benchmark's selective-refusal contract.

Ablation



Manifest: outputs/paper-facing-manifest.json | LLM matrix: Full_Llm_baseline_matrix-completed-Poster generated from frozen JSON artifacts

OOD Stress



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Paired-flip is the clearest current failure slice: acc 0.439.

Graph/mechanism buckets are saturated in this small snapshot.

Validity Gates

Leakage control: oracle-positive reaches 1.000/1.000; score gain is contamination.

LLM matrix: 5 models, n=1245, failed=0, parse=0, fallback=0; best OOD GPT-5.5 acc=0.545,

Human audit: dual annotation pending; no real-grounded headline.